

```
import pandas as pd

# Correct dataset URL
url = "https://raw.githubusercontent.com/shivang98/Social-Network-ads-Boost/refs/heads/master/Social_Network_Ads.csv"

# Load dataset
dataset = pd.read_csv(url)

# Preview
print(dataset.head())
```

	User ID	Gender	Age	EstimatedSalary	Purchased
0	15624510	Male	19	19000	0
1	15810944	Male	35	20000	0
2	15668575	Female	26	43000	0
3	15603246	Female	27	57000	0
4	15804002	Male	19	76000	0

```
print(dataset.columns)
print(dataset.shape)
```

```
Index(['User ID', 'Gender', 'Age', 'EstimatedSalary', 'Purchased'], dtype='object')
(400, 5)
```

```
dataset = dataset[['Age', 'EstimatedSalary', 'Purchased']]
```

```
X = dataset[['Age', 'EstimatedSalary']].values
y = dataset['Purchased'].values
```

```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, random_state=0
)
```

```
from sklearn.preprocessing import StandardScaler
sc = StandardScaler()
X_train = sc.fit_transform(X_train)
X_test = sc.transform(X_test)
```

```
from sklearn.linear_model import LogisticRegression
model = LogisticRegression()
model.fit(X_train, y_train)
```

▼ LogisticRegression ⓘ ?

```
LogisticRegression()
```

```
y_pred = model.predict(X_test)

print("\nPredicted values:\n", y_pred[:10])
```

```
Predicted values:
[0 0 0 0 0 0 0 1 0 1]
```

```
from sklearn.metrics import confusion_matrix
cm = confusion_matrix(y_test, y_pred)

print("\nConfusion Matrix:\n", cm)
```

```
Confusion Matrix:
[[65  3]
 [ 8 24]]
```

```
TN = cm[0][0]
FP = cm[0][1]
FN = cm[1][0]
TP = cm[1][1]
```

```
accuracy = (TP + TN) / (TP + TN + FP + FN)
error_rate = (FP + FN) / (TP + TN + FP + FN)
```

```
precision = TP / (TP + FP)
recall = TP / (TP + FN)
```

```
print("\nPerformance Metrics:")
print("Accuracy:", accuracy)
print("Error Rate:", error_rate)
print("Precision:", precision)
print("Recall:", recall)
```

```
Performance Metrics:
Accuracy: 0.89
Error Rate: 0.11
Precision: 0.8888888888888888
Recall: 0.75
```

```
import matplotlib.pyplot as plt
import numpy as np
from sklearn.metrics import confusion_matrix

# Assuming you already have y_test and y_pred
cm = confusion_matrix(y_test, y_pred)

# Plot
plt.figure()
plt.imshow(cm)

# Labels
plt.title("Confusion Matrix")
plt.xlabel("Predicted Label")
plt.ylabel("True Label")

# Tick marks
classes = ['No Purchase', 'Purchase']
plt.xticks(np.arange(len(classes)), classes)
plt.yticks(np.arange(len(classes)), classes)

# Add numbers inside boxes
for i in range(len(cm)):
    for j in range(len(cm)):
        plt.text(j, i, cm[i][j], ha="center", va="center")

plt.colorbar()
plt.show()
```



